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Summary

Computer Science and Business student working the full stack of a machine learning product: research and training, the services that run inference, and the front ends on top. Kaggle top 2%; runs a graph neural network on live market data.

Experience

Machine Learning Intern

Jun 15 – Jul 21, 2026

KeyPoint Technologies (Xplore)

- Owned the language model and the iOS keyboard extension end to end. Trained and ablated a family of **4M–11M parameter** models as a drop-in replacement for the shipped autocorrection and next-word prediction engine, then carried the winning configuration through **Core ML** conversion, int8 quantization, and on-device latency profiling.
- Held the extension inside the **66MB** memory ceiling iOS enforces on keyboard extensions, cutting the resident baseline to **15MB** by memory-mapping the 60k-word dictionary trie and n-gram tables as clean, evictable pages. Decoding runs a trie-constrained beam search pruned against the model's next-token distribution.

Software Developer

Dec 2022 – Mar 2023

VHR Infra

- Sole engineer on the company's production site: **Next.js** front end on **Firebase**, from data model through deploy.
- Built a logistic regression model (scikit-learn) scoring whether a prospect would accept a quoted price, and prototyped a house-price regression on client transaction data.

Founding Member

Dec 2021 – Jun 2022

Y2C (student-founded NGO)

- Sole developer of the organization's website: **Next.js** with server-side rendering, deployed to **Vercel**, **Firebase** for persistence and team auth.
- Handled all technology for the organization, from architecture through deployment, and represented it to Y Combinator contacts and NGO backers.

Projects and Competitions

Graph Neural Network Intraday Options Trading — Live Portfolio

Graph neural network that jointly embeds a universe of equities and learns their dependencies, driving real-time intraday options execution through the **Zerodha Kite API**. Live portfolio; peak single-day return **220%**, max drawdown 60%.

Triver — Personal Project

2021 – Present

Preference-first travel recommender spanning **SVD++**, **Seq2Seq**, and **graph neural network** retrieval, served behind an automated training-to-deployment pipeline with inference optimization and production monitoring.

Reinforcement Learning — Implementations

Implemented World Models (VAE, MDN-RNN, CMA-ES controller) on CarRacing, learning a policy inside the model's own latent rollouts; also **PPO**, **SAC**, **DDPG**, and ICM curiosity-driven exploration from the original papers, plus GANs for generative modelling.

Stable Diffusion Image-to-Prompts — Kaggle

Top 2% of a \$50,000 competition. Fine-tuned a CLIP image encoder with an Efficient Channel Attention head to regress prompt embeddings under a cosine-similarity objective, and evaluated Q-Former variants against it.

BirdCLEF 2026 — Kaggle

Top 10% of a \$50,000 competition. Constrained the embedding space to an equiangular tight frame (ETF), borrowing the neural-collapse geometry to hold class separation for few-sample bird species in field audio.

ConserVision Wildlife Image Classification

15th place. Species classification robust to camera-trap artifacts: low light, partial occlusion, and motion-triggered crops.

Skills

Languages: Python, Java, Swift, Kotlin, Go, JavaScript / TypeScript, PHP, MATLAB, CUDA

Machine Learning: PyTorch, TensorFlow, JAX, Flax, scikit-learn, XGBoost, Hugging Face Transformers

Training and Inference: LoRA / PEFT, SFT and DPO, DeepSpeed, FSDP, Ray, vLLM, ONNX Runtime, TensorRT, Core ML, quantization, distillation

MLOps: Weights & Biases, MLflow, Airflow, Docker, Kubernetes

Web and Mobile: React, React Native, Next.js, SwiftUI, micro frontends, Firebase, Vercel

Systems and Data: microservices, Kafka, NATS, Ejabberd, PostgreSQL, TimescaleDB, Elasticsearch, MongoDB, Redis

Education

Two concurrent undergraduate degrees, permitted under UGC's 2022 dual-degree policy (one online, one on campus).

B.Sc. Computer Science — Birla Institute of Technology and Science (BITS), Pilani, Rajasthan 2025 – 2028
Online Degree Programme.

BBA, Artificial Intelligence and Data Science — ICFAI, Hyderabad; full-time, on campus

2024 – 2027

High School, Indus International School, Hyderabad

Grade 10, Delhi Public School, Hyderabad